

Raquel Sweet

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EDUCATION

Cornell University, College of Engineering, Ithaca NY - **GPA: 4.02** **B.S. Expected May 2027**
Bachelor of Science, Mechanical Engineering; Dyson Business Minor for Engineers
Member of Tau Beta Pi engineering honors society (top 12 % engineering class) and Sigma Delta Tau Sorority

PROFESSIONAL EXPERIENCE

Boom Supersonic, Engine Systems Intern **June 2026 - Aug 2026**

- Designed fluid distribution networks for water coolant, compressed air, nitrogen, and hydrogen using Visio and Onshape, supplying 4 skids and a blade tip timing cooling system for Sprint Core, Boom's supersonic engine demonstrator
- Met safety specifications for each skid while working within the constraints of a single shared air compressor and water chiller
- Hand-routed fluid tubing to fixed skid locations, navigating building and electrical constraints to avoid interference
- Designed and built weather station to capture real-time atmospheric data, engineering it to withstand winds up to 115 mph

Magnus Ultralight Research Project, Cornell University, *Undergraduate Researcher* **May 2025 - Dec 2025**

- Contributed to the build and testing of the first production Magnus effect-powered ultralight aircraft
- Fabricated prototype of eVTOL aircraft, including quadcopter rotor system, landing skids, and wiring system
- Flight-tested aircraft lifting 4 kg of external payload for the duration of ten five-minute hovers
- Applied data from ANSYS and MATLAB to guide design decisions
- Developed models using SOLIDWORKS and Fusion 360 for aircraft design and to aid in analysis
- Designed and built a gear-driven hub system for rotating a Magnus Effect sphere at over 1,500 RPM, ensuring dynamic balance for smooth and stable operation

Design Build Fly Project Team, Cornell University, *Mechanical Sub-team Lead/Composite Lead* **Oct 2023 - Present**

- Lead a 12-member sub-team responsible for the mechanical and structural subsystems of an aircraft to compete in the international AIAA DBF competition for the 2025-26 year
- Utilize FEA to lead structural analysis of mechanical subsystems, including landing gear and fuselage
- Lead dynamic testing of aircraft towable banner to validate aerodynamic drag models and optimize design performance
- Supervise and support sub-team projects, and provide mentorship and onboarding for new members
- Developed DBF's first ever composite fabrication plan by gauging material feasibility and tensile strength and researching available resources
- Successfully implemented plan and reduced fuselage mass by 65% from previous year with carbon fiber model
- Designed fuselage of aircraft utilizing CAD software and manufactured double layered carbon fiber fuselage by hand cutting and layering components

WORK EXPERIENCE

Statics and Mechanics of Solids Grader, Cornell University **Aug 2026 - Present**

- Grade weekly assignments for Statics and Mechanics of Solids ensuring consistent evaluation across 300 students

Academic Excellence Workshop Facilitator: Statics, Cornell University **Aug 2025 - Dec 2025**

- Taught 20 students in a weekly supplementary class for the Statics and Mechanics of Solids course
- Prepared original presentations and review material for each session
- Attended eight trainings to improve lesson plans, teaching techniques, and conflict resolution

CAMPUS INVOLVEMENT

HumorUs - Cornell University Sketch Comedy Group **Oct 2024 - Present**

- Write and revise sketch material, incorporate peer feedback, and perform live for audiences of 400 at semi-annual showcases

SPECIALIZED SKILLS

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- Software:** SOLIDWORKS, Fusion 360, Onshape, Python, MATLAB, ANSYS (Fluent and Static Structural)
 - Hardware:** Composite Fabrication, Machining (bandsaw, lathe, and mill), 3D printing, Soldering